

ACN Practical 4

1. Define Routing table.

Ans. A routing table is a data table stored in a router that contains information about available network paths. It helps the router determine where to forward incoming data packets so they can reach their destination.

A routing table typically contains:

- Destination network
- Subnet mask/prefix
- Next-hop address
- Outgoing interface
- Route metric/cost

2. State the purpose of queuing in routers.

Ans. Queuing in routers is used to temporarily store packets when the outgoing link is busy or congested.

It helps the router:

- Manage network traffic efficiently.
- Prevent packet loss during congestion.
- Decide the order in which packets are transmitted.
- Handle situations where packets arrive faster than they can be sent.

3. Distinguish between dynamic routing and static routing on the basis of configuration, security, cost and routing protocols.

Ans.

Basis	Static Routing	Dynamic Routing
Configuration	Routes are manually configured by the network administrator.	Routes are automatically learned and updated by routing protocols.
Security	More secure because routes are not exchanged with other routers.	Comparatively less secure because routing information is exchanged between routers.
Cost	Low cost because it requires fewer router resources and no routing protocol overhead.	Higher cost due to CPU, memory, and bandwidth used by routing protocols.
Routing Protocols	Does not use dynamic routing protocols.	Uses protocols such as RIP, OSPF, EIGRP, and BGP .

4. Define:

i) Inter-domain routing

ii) Intra domain routing

Ans. i) Inter-domain Routing (IDR):

Inter-domain routing is the process of routing data between different autonomous systems (AS) or separate administrative networks on the Internet.

Example: BGP (Border Gateway Protocol).

ii) Intra-domain Routing:

Intra-domain routing is the process of routing data within a single autonomous system (AS) or one administrative network.

Examples: RIP, OSPF, and EIGRP.

5. Compare Distance Vector Routing and Link State Routing.

Ans.

Basis	Distance Vector Routing	Link State Routing
Routing Information	Shares information about routes with neighboring routers.	Shares information about the state of its directly connected links.
Knowledge of Network	Has limited knowledge of the network topology.	Has complete knowledge of the network topology.
Algorithm Used	Uses Bellman-Ford algorithm.	Uses Dijkstra's Shortest Path First (SPF) algorithm.
Route Calculation	Calculates routes based on distance and direction from neighbors.	Calculates the shortest path using the complete network topology.
Convergence	Generally slower.	Generally faster.
Resource Usage	Requires less CPU and memory.	Requires more CPU and memory.
Examples	RIP	OSPF, IS-IS

6. Describe the RIP message format along with its features.

Ans. RIP (Routing Information Protocol) is a distance-vector routing protocol used to exchange routing information between routers within an autonomous system.

RIP Message Format

A RIP message consists of a header followed by one or more Route Entries.

Field	Description
Command	Specifies the type of message: Request (1) or Response (2).
Version	Indicates the RIP version, such as RIPv1 or RIPv2.
Must be Zero	Reserved field that is normally set to zero.
Address Family Identifier (AFI)	Identifies the network address type, usually IPv4.
IP Address	Specifies the destination network address.
Subnet Mask	Specifies the subnet mask of the destination network (used in RIPv2).
Next Hop	Specifies the next-hop router for the destination (used in RIPv2).
Metric	Indicates the number of hops required to reach the destination.

Features of RIP

1. **Distance-vector protocol** – Uses distance (hop count) to select routes.
2. **Hop count metric** – Maximum hop count is 15; 16 means unreachable.
3. **Periodic updates** – Routers normally send routing updates every 30 seconds.
4. **Simple to configure** – Suitable for small networks.
5. **Slow convergence** – It may take time to adapt to network changes.
6. **RIPv1** – Classful protocol and does not include subnet mask information.
7. **RIPv2** – Classless protocol and supports VLSM and CIDR.
8. **Administrative Distance** – RIP has an administrative distance of 120.